

# EXPLANATIONS

1. (d) Heat capacity ( $C$ ) =  $\frac{\Delta Q}{\Delta T}$ , unit:  $\text{J K}^{-1}$

Specific heat capacity ( $s$ ) =  $\frac{\Delta Q}{m\Delta T}$ ,

unit:  $\text{J kg}^{-1}\text{K}^{-1}$ .

Latent heat ( $L$ ) =  $\frac{\Delta Q}{m}$ , unit:  $\text{J kg}^{-1}$ .

Thermal conductivity ( $K$ ) from  $\Delta Q = -KA \frac{\Delta T}{\Delta x}$ , unit:  $\text{J m}^{-1}\text{K}^{-1}\text{s}^{-1}$

Thus, match: (A)-(II), (B)-(III), (C)-(I), (D)-(IV).

2. (\*) DROPPED by NTA

Magnetic intensity

$$\left(H = \frac{B}{\mu}\right) \rightarrow \text{Ampere/meter}$$

Magnetic moment ( $\vec{M} = i\vec{A}$ )

**Note:** None of the option(s) are correct but if we need to choose most appropriate option then the answer is (d)

3. (b)  $\vec{E} = \frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}$

$$\left[\frac{A}{x^2}\right] = NC^{-1} \Rightarrow [A] = Nm^2C^{-1}$$

$$\left[\frac{B}{y^3}\right] = NC^{-1} \Rightarrow [B] = Nm^3C^{-1}$$

4. (c) Unit of  $R_H$  :  $\text{m}^{-1}$ ,  $h$  :  $\text{kg m}^2\text{s}^{-1}$ ,  $\mu_B$  :  $\text{kg m}^{-1}\text{s}^{-2}$ ,  $\eta$  :  $\text{kg m}^{-1}\text{s}^{-1}$ ,

5. (c)  $\frac{E}{H} = \frac{\text{Volt/metre}}{\text{Ampere/metre}} = \frac{\text{Volt}}{\text{Ampere}} = \text{ohm}$

6. (a)  $128[M_1L_1^{-3}] = n_2[M_2L_2^{-3}]$   
 $\Rightarrow n_2 = 128 \left[ \frac{M_1}{M_2} \right] \left[ \frac{L_1}{L_2} \right]^{-3}$   
 $= 128 \left[ \frac{1000}{50} \right] \left[ \frac{100}{25} \right]^{-3} = \frac{128 \times 20}{64} = 40$

7. (d) Inductance  $L$  and capacitance  $C$  relate to  $\mu_0$  and  $\epsilon_0$ , with  $C = \frac{A\epsilon_0}{d}$  and  $L = \mu_0 nI$ . The ratio  $\frac{L}{C}$  equals  $(R)^2$ , where  $R$  is resistance. Therefore,  $\sqrt{\frac{\mu_0}{\epsilon_0}}$  has the dimension of  $R$ , the answer is (d).

8. (c)

(A) Mass density =  $\frac{M}{V}$ , unit =  $[\text{ML}^{-3}]$ .

(B) Impulse =  $M.u$ , unit =  $[\text{MLT}^{-1}]$ ,

(C) Power =  $F.V$ , unit =  $[\text{ML}^2\text{T}^{-3}]$ ,

(D) Moment of inertia =  $Mr^2$ , unit =  $[\text{ML}^2]$

9. (a)

(A)  $[k] = \frac{PV}{NT} = \frac{ML^2T^{-2}}{K} = \text{ML}^2T^{-2}K^{-1}$

(B)  $[\eta] = \frac{F}{6\pi\eta v} = \frac{MLT^{-2}}{L^2T^{-1}} = \text{ML}^{-1}T^{-1}$

(C)  $[h] = \frac{E}{f} = \frac{ML^2T^{-2}}{T^{-1}} = \text{ML}^2T^{-1}$

(D)  $\frac{dQ}{dt} = k \frac{AdT}{dx}$   
 $k = \frac{(ML^2T^{-2})L}{L^2.K} = \text{MLT}^{-3}K^{-1}$

10. (b)  $[q] = [\text{AT}]$

$[\mu_0] = [\text{M LT}^{-2}\text{A}^{-2}]$

Momentum  $[P] = [\text{MLT}^{-1}]$

$[qm_0I] = [\text{AT}][\text{MLT}^{-2}\text{A}^{-2}][\text{AT}] = \text{MLT}^{-1}$ , matching momentum dimensions.

11. (b) The speed of light

$$C = \frac{1}{\sqrt{\mu_0\epsilon_0}} \Rightarrow \frac{1}{\mu_0\epsilon_0} = C^2.$$

Since  $C$  has dimensions

$[\text{LT}^{-1}]$ ,  $C^2 = [\text{L}^2\text{T}^{-2}]$ .

Thus, the dimension is  $\text{L}^2\text{T}^{-2}$ .

- (a)  $\left[ \frac{B}{\mu_0} \right] = [n] = L^{-1} A$
- (c) Capacitance is defined as:
- $$C = \frac{Q}{V}$$
- Charge  $Q$  has dimensional formula  $[Q] = [C]$
  - Voltage  $V$  is energy per charge:
- $$V = \frac{W}{Q} = \frac{ML^2 T^{-2}}{C} = ML^2 T^{-2} C^{-1}$$
- Now, substituting in  $C = \frac{Q}{V}$ :
- $$[C] = \frac{C}{ML^2 T^{-2} C^{-1}} = C^2 M^{-1} L^{-2} T^2$$
- (d)  $x(t) = A \sin(\omega t) + B \cos(\omega t) + C t^2 + D$   
Here, A, B, C, and D are constants, and  $t$  is time.
- $$[A] = L, [B] = L, [C] = \frac{L}{T^2}, [D] = L$$
- $$\left[ \frac{ABC}{D} \right] = \left[ \frac{L \times L \times L T^{-2}}{L} \right] = [L^2 T^{-2}]$$
- (b) (D) Torsional constant  $C = C \theta$   
 $C = [MLT^{-2}]$   
 (C) Magnetic moment  $M = NIA$   
 $M = [L^2 A]$   
 $(B) F = qVB$   
 $[MLT^{-1}] \cdot [AT] [LT^{-1}] B$   
 Magnetic field  
 $B = [MT^{-1} A^{-1}]$
- (A)  $B = \frac{\mu_0 I}{2r}$
- Permeability of free space
- $$\mu_0 = \frac{[MT^{-1} A^{-1}] [L]}{[A]}$$
- $$[\mu_0] = [MLT^{-2} A^{-2}]$$
- (c)  $[E] = M L^2 T^{-2}$   
 $[P] = M L^{-1} T^{-1} L^2 T^{-1} = M^2 L T^{-2}$   
 $[mc^2] = M L^2 T^{-2} \Rightarrow [E] = [mc^2]$   
 From principle of homogeneity  
 $E^2 = P^2 c^2 = m^2 c^4$
17. (a)  $\left[ \frac{\text{Modulus of elasticity}}{\text{TORQUE}} \right]$   
 $= \left[ \frac{\text{STRESS}}{\text{STRAIN} \cdot \text{TORQUE}} \right] = [L]$   
 hence  $C = 0$
18. (c) Angular impulse has dimension  $[ML^2 T^{-1}]$   
 Latent Heat has dimension  $[M^2 L^2 T^{-2}]$   
 Electrical resistivity has dimension  $[ML^2 T^{-1} A^{-1}]$

Electromotive force has dimension

$$[ML^2 T^{-1} A^{-1}]$$

19. (b) The dimensional formula of surface tension =  $[MT^{-1}]$ , impulse =  $[MLT^{-1}]$  which does not match.

$$20. (c) [LT^{-1}] = [A] \left[ \frac{B}{C+T} \right]$$

$$\begin{aligned} [C] &= [T] \\ [A] &= [LT^{-1}] \\ [B] &= [LT^{-1}] \\ [ABC] &= [L^2 T^{-1}] \end{aligned}$$

$$21. (a) (A) \left[ \gamma \right] = \frac{F}{A \left( \frac{\Delta T}{L} \right)} = ML^2 T^{-2} \left( \frac{L}{\Delta T} \right) = \text{Dimensionless}$$

(B) Torque

$$(\tau) = \vec{r} \times \vec{F} = L \times MLT^{-2} = ML^2 T^{-2}$$

(C) Coefficient of viscosity  $\Rightarrow \frac{F}{A \cdot \frac{dv}{dy}}$   
 $\eta \rightarrow Pa \text{ sec.} \quad (\text{Pascal} - \text{sec.})$

$$[\eta] = \frac{MLT^{-2}}{L^2} \times T = ML^{-1} T^{-1}$$

(D) Gravitational constant (G)

$$[G] = \frac{F \cdot r^2}{m_1 m_2} = \frac{MLT^{-2} \times L^2}{M^2} = M^{-1} L^3 T^{-2}$$

22. (c) Heat =  $m L \Rightarrow L = \text{heat}/m$   
 $\Rightarrow [L] = \frac{MLT^{-2}}{M} = M^0 L^3 T^{-2}$

23. (b) [Energy Density] =  $\frac{1}{2} [\epsilon_0 E^2]$   
 $[\epsilon_0 E^2] = \frac{[\text{Energy}]}{[\text{Vol.}]} = \frac{ML^2 T^{-2}}{L^3}$   
 $= [ML^{-1} T^{-2}]$

24. (c)  $\alpha = \frac{\Delta Q}{m \Delta T}$   
 $[\alpha] = \frac{ML^2 T^{-2}}{MK}$   
 $[\alpha] = [L^2 T^{-2} K^{-1}]$

Statement-I is correct

$$PV = nRT \Rightarrow R = \frac{PV}{nT}$$

$$[R] = \frac{[ML^{-1} T^{-2}] [L^3]}{[mol] [K]}$$

$$[R] = [ML^2 T^{-2} mol^{-1} K^{-1}]$$

Statement-II is incorrect

25. (b)  $[F] = [\vec{r} \times \vec{F}] = [ML^2 T^{-2}]$   
 $\Rightarrow B = \left( \frac{F}{qV} \right) = \left[ \frac{MLT^{-2}}{ATLT^{-1}} \right] = [MA^{-1} T^{-1}]$

$$[M] = [L \times A] = [AL^2]$$

$$B = \frac{\mu_0 I d \sin \theta}{4\pi r^2}$$

$$\Rightarrow [\mu] = \left[ \frac{B r^2}{I d} \right] = \left[ \frac{MT^{-2} A^{-1} L^2}{AL} \right]$$

$$= [MLT^{-1} A^{-1}]$$

26. (b)  $[b] = [\text{Vol.}] = [L^3]$

$$\left[ \frac{a}{V} \right] = [P] \Leftrightarrow [a] = [VP] = [MLT^{-2}]$$

$$[ab^{-1}] = [MLT^{-2}]$$

27. (b) Here, we have  $G$  is gravitational constant and  $u$  is the energy density.

$$[uG] = [(ML^{-1} T^{-2}) (M^2 L^2 T^{-4})] = [ML^2 T^{-4}]$$

$$\text{So, } [\sqrt{uG}] = [L^2 T^{-2}]$$

28. (d) Change in angular momentum = Angular impulse

$$[\text{Angular momentum}] = [\text{Angular impulse}]$$

$$\Rightarrow [mvr] = [M L^2 T^{-1}]$$

29. (a)  $[P] = MLT^{-1}$   
 $[v] = MLT^{-1}$   
 $[h] = ML^2 T^{-1}$   
 $[L] = ML^2 T^{-1}$

(Here  $P$  is linear momentum and  $\tau$  is moment of force,  $h$  is Planck's constant,  $L$  is angular momentum).

30. (c) As we know,

$$F = \eta A \frac{dv}{dy}, L = mvr, S.T. = \frac{F}{\ell}$$

$$K.E. = \frac{1}{2} I \omega^2$$

$$\text{Hence, } F = \eta A \frac{dv}{dy}$$

$$[MLT^{-2}] = \eta [L^2] [T^{-1}]$$

$$\eta = [ML^{-1} T^{-1}]$$

$$S.T. = \frac{F}{\ell} = \frac{[MLT^{-2}]}{[L]} = [ML^0 T^{-2}]$$

$$L = mvr = [MLT^{-1}]$$

$$K.E. = \frac{1}{2} I \omega^2 = [ML^2 T^{-2}]$$

31. (b)  $\gamma = \frac{\text{Stress}}{\text{Strain}} = \left[ \frac{MLT^{-2}}{L^2} \right] = [ML^{-1} T^{-2}]$   
 $\therefore \eta = \frac{F}{6\pi r v}$   
 $\Rightarrow [\eta] = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1} T^{-1}]$   
 $\therefore h = \frac{E}{v} \Rightarrow h = \frac{[ML^2 T^{-2}]}{[T^{-1}]} = [ML^2 T^{-1}]$

Work function has same dimension as that of energy, so  $[\phi] = [ML^2 T^{-2}]$

32. (c) Pressure gradient  
 $= \frac{[\text{Pressure}]}{[\text{Length}]} = \frac{[ML^{-1} T^{-2}]}{[L]} = [ML^{-2} T^{-2}]$

Energy density =  $\frac{\text{energy}}{\text{volume}} = [ML^{-1} T^{-2}]$

Electric field =  $\frac{\text{Force}}{\text{charge}} = \frac{[MLT^{-2}]}{[AT]}$

$$= [MLT^{-1} A^{-1}]$$

Latent heat =  $\frac{\text{heat energy}}{\text{mass}} = \frac{[ML^2 T^{-2}]}{[M]}$

$$= [ML^2 T^{-2}]$$

33. (b) (A) Surface Tension

$$= \frac{F}{\ell} = \frac{MLT^{-2}}{L} = MT^{-2}$$

$$= Kgs^{-2} \Rightarrow \text{option IV}$$

(B) Pressure =  $\frac{F}{A} = \frac{MLT^{-2}}{L^2}$

$$= Kgm^{-1} s^{-2} \Rightarrow \text{option (III)}$$

(C) viscosity =  $\frac{F}{A \left( \frac{dv}{dy} \right)} = \frac{MLT^{-2}}{L^2 \left( \frac{LT^{-1}}{L} \right)}$

$$= ML^{-1} T^{-1}$$

$$= Kgm^{-1} s^{-1} \Rightarrow \text{option (I)}$$

(D) impulse =  $\int F dt = MLT^{-2} \times T$

$$\Rightarrow (A) - (IV), (B) - (III), (C) - (I), (D) - (II)$$

34. (a) Spring Constant

$$[K] = \frac{[F]}{[X]} = \frac{MLT^{-2}}{L} = MT^{-2}$$

$$\text{Angular speed } [\omega] = \frac{[\theta]}{[t]} = \frac{1}{T} = T^{-1}$$

Angular Momentum,

$$[L] = [M] [V] \cdot [R] = [M] [LT^{-1}] [L] = [ML^2 T^{-1}]$$

Moment of Inertia,

$$[I] = [M] [R]^2 = [M] [L^2] = [ML^2]$$

35. (b) [Stress]

$$= \frac{[F]}{[A]} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1} T^{-2}]$$

[Pressure gradient]

$$= \frac{[P]}{[Z]} = \frac{[ML^{-1} T^{-2}]}{[L]} = [ML^{-2} T^{-2}]$$

$$[\text{Top}\theta\text{v}] = [r][F] = [L][MLT^{-2}] = [ML^2 T^{-2}]$$

$$[\text{Angular Momentum}] = [\tau][\ell] = [ML^2 T^{-2}]$$

36. (d)

$$\therefore c^2 = \frac{1}{\mu_0 \epsilon_0} \Rightarrow \left[ \frac{1}{\mu_0 \epsilon_0} \right] = [L^2 T^{-2}]$$

38. (b) (A) Planck's constant

$$h = \frac{E}{\nu} = \frac{[M^1 L^2 T^{-2}]}{[T^{-1}]} = [M^1 L^2 T^{-1}]$$

$$(B) E = qV$$

$$V = \frac{E}{q} = \frac{[ML^2 T^{-2}]}{[AT]} = [ML^2 T^{-1} A^{-1}]$$

(C)  $\phi$  (work function)

$$= \text{energy} = [ML^2 T^{-2}]$$

(D) Momentum ( $p$ ) =  $F \cdot t = [MLT^{-2} T] = [MLT^{-1}]$

39. (a) We know that,

$$\text{Velocity gradient} = \frac{dv}{dx}$$

$$\text{Decay constant } (\lambda) = \frac{0.693}{T_{1/2}}$$

$$\therefore \text{Dimension of velocity gradient}$$

$$\left( \frac{dv}{dx} \right) = \frac{LT^{-1}}{L} = T^{-1}$$

$$40. (c) e - m \frac{dv}{dt} \Rightarrow [m] = \left[ \frac{e m f}{\frac{dv}{dt}} \right]$$

$$= \frac{[ML^2 T^{-2} A^{-1} T^{-1}]}{[AT^{-1}]} = [ML^2 T^{-2} A^{-2}]$$

$$41. (a) [\text{Pascal-second}] = \frac{[MLT^{-2}]}{[L^2]}$$

$$= [ML^{-1} T^{-1}]$$

42. (a) Unit of Viscosity = Pascal Second.

$$\text{Therefore, dimension of viscosity} = [ML^{-1} T^{-1}]$$

According to question,

$$[ML^{-1} T^{-1}] = P^x A^y T^z$$

$$[ML^{-1} T^{-1}] [L^2] [T] = [M^x L^y T^z]$$

$$[M^1 L^{-1+2} T^{-1+1}] = [M^x L^y T^z]$$

$$x = 1, x + 2y = -1, -x + z = -1$$

$$y = -1, z = 0$$

$$\text{Viscosity} = [P^1 A^{-1} T^0]$$

$$43. (c) \frac{B^2}{\mu_0} \text{ is energy density in magnetic field.}$$

$$\therefore \left[ \frac{B^2}{\mu_0} \right] = [\text{Energy density}]$$

$$= [\text{Energy}]/[\text{Volume}]$$

$$= \frac{[ML^2 T^{-2}]}{[L^3]} = [ML^{-1} T^{-2}]$$

$$44. (a) \text{ Electric Displacement, } \vec{D} = \epsilon_0 \vec{E}$$

From Gauss's Law,

$$EdA = \frac{q}{\epsilon_0} \text{ or } [\epsilon_0 E] = \left[ \frac{q}{dA} \right] = [AT L^{-2}]$$

$$\text{Surface charge density } (\sigma)$$

$$\sigma = \frac{Q}{A} \Rightarrow [\sigma] = \frac{[Q]}{[A]} = \frac{[AT]}{[L^2]} = [LT^{-1} A]$$

$$\text{Electric displacement } (\vec{D}) \text{ and surface charge density } (\sigma) \text{ have the same dimension.}$$

45. (b) Torque =  $\vec{r} \times \vec{F}$  (Nm)

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \quad (N/m^2)$$

$$\text{Latent heat} = \frac{\text{Energy}}{\text{Mass}} \quad (J kg^{-1})$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \quad (N ms^{-1})$$

$$46. (d) \Delta Q = ms \Delta T \Rightarrow [s] = \frac{[\Delta Q]}{[m][\Delta T]}$$

$$\Rightarrow [s] = \frac{[ML^2 T^{-2}]}{[M][L]} = L^2 T^{-2} K^{-1}$$

$$\Delta Q = mL \Rightarrow [L] = \frac{ML^2 T^{-2}}{M} = L^2 T^{-2}$$

$$47. (a) \text{ Capacitance}$$

$$[C] = \frac{Q}{V} = \frac{Q}{W/Q} = \frac{Q^2}{W} = \frac{A^2 T^2}{M^2 L^2 T^{-2}}$$

$$= M^{-1} L^2 T^4 A^2$$

$$\text{Permittivity } [\epsilon_0] = M^{-1} L^3 T^4 A^2$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow \mu_0 = \frac{1}{\epsilon_0 c^2}$$

$$\Rightarrow [\mu_0] = \frac{[ML^{-1} T^{-2}]}{[L^2]} = [ML^{-3} T^{-2}]$$

$$\text{Electric field, } E = \frac{F}{q} = \frac{MLT^{-2}}{AT}$$

$$\Rightarrow [E] = MLT^{-1} A^{-1}$$

48. (a)  $P = EL^2 M^{-1} G^{-2}$   
 $= [ML^2 T^{-2}] [ML^2 T^{-2}]^2 [M^1 L^{-1} T^{-1}]^2$   
 $= M^5 L^7 T^{-7}$

49. (d)  $\lambda = \frac{c}{\nu} = \frac{Q/V}{V} = \frac{Q}{V^2} = \frac{Q^2}{W^2} = \frac{J^2 T^4}{F^2 \cdot J^2}$

$$\Rightarrow [\lambda] = \frac{[L^2 T^{-2}]}{[M^1 L^2 T^{-1}]} = [M^{-1} L^0 T^{-1}]$$

50. (d) The dimension of  $[\mu_0]$   
 $= [ML^2 T^{-1} A^{-2}]$

$$\therefore \mu_0 = \frac{B_0}{H} = NA^{-2}$$

It is not dimensionless

51. (c)  $[h] = [ML^2 T^{-1}]$   
 $[E] = [ML^2 T^{-2}]$   
 $[P] = [ML^2 T^{-1} L^{-1}]$   
 $[P] = [MLT^{-1}]$

52. (c) We know that,  $[F] = MLT^{-2}$   
 $\Rightarrow M = FL T^2$

$$[p] = ML^{-1} = FL^{-1} T^2 L^{-1} = FL^{-2} T^2$$

53. (c)

$$(a) \text{ Torque} \rightarrow ML^2 T^{-2} \rightarrow (iii)$$

$$(b) \text{ Impulse} \rightarrow MLT^{-1} \rightarrow (i)$$

$$(c) \text{ Tension} \rightarrow MLT^{-2} \rightarrow (iv)$$

$$(d) \text{ surface Tension}$$

$$\rightarrow \frac{MLT^{-2}}{L} = MT^{-2} \rightarrow (ii)$$

54. (b)  $p = k S^{-1} t^k$   
where  $k$  is dimensionless constant  
 $MLT^{-1} = (MT^{-2})^k (ML^2 T^{-1})^k$   
 $a + b + c = 1$  ... (i)  
 $2b + 2c = 1$  ... (ii)  
 $-2a - c = 1$  ... (iii)

On solving the above equations

$$a = \frac{1}{2}, b = \frac{1}{2}, c = 0$$

Thus,  $p \propto S^{1/2} t^{1/2}$

55. (d) Dimension for permittivity  $[\epsilon_0] = M^{-1} L^{-1} T^2 A^2$   
Dimension for permeability  $[\mu_0] = M L T^{-2} A^{-2}$   
Dimension for electrical resistance  $[R] = M L^2 T^{-3} A^{-2}$   
Using the given options

$$[R] = \left[ \frac{V}{I} \right] = \frac{MLT^{-2}A^{-1}}{AT^{-1}} = ML^2T^{-3}A^{-2}$$

56. (c) Dimension of

$$\left[ \frac{\epsilon_0}{\mu_0} \right] = \left[ \frac{M^{-1} L^{-1} T^2 A^2}{M L T^{-2} A^{-2}} \right] = [M^{-1} L^{-2} T^4 A^4]$$

57. (d) The real gas equation is  $\left( P + \frac{a}{V^2} \right) (V - b) = RT$ , where  $P, V, T$ , and  $R$  are pressure, volume, temperature, and gas constant.

To find the dimension of  $ab^2$ , note that  $a/V^2$  has dimensions of pressure (since it adds to  $P$ ).

$$\text{so } [a] = [P][V^2] = ML^{-1}T^{-2}L^4 = ML^3T^{-2}$$

$$\text{Since } [b] = [V] = [L]^3$$

$$\therefore [ab^2] = [ML^3T^{-2}][L^3]^2 = [ML^9T^{-2}]$$

58. (b) The electric flux  $\phi$  has dimensions of charge per unit area, i.e.,

$$[\phi] = \left[ \frac{Q}{\text{Area}} \right]$$

$\sigma$  is associated with surface charge density which has the dimensions:

$$[\sigma] = \frac{Q}{\text{Area}}$$

Thus, the dimensions of  $\sigma$  will be:

$$[\sigma] = \left[ \frac{Q}{A} \right] = \frac{Q/L^2}{Q/L^2} = 1$$

Next, for linear charge density, we have:

$$[\lambda] = \frac{Q}{\text{Length}}$$

$$[\rho] = \left[ \frac{Q}{A} \right] = \frac{Q/L^2}{Q/L^2} = \frac{\text{Length}}{\text{Area}}$$

Now, considering the ratio  $\frac{\sigma}{\rho}$ , we find:

$$\left[ \frac{\sigma}{\rho} \right] = \frac{1}{\frac{\text{Length}}{\text{Area}}} = \text{Length}$$

Thus, the ratio  $\left[ \frac{\sigma}{\rho} \right]$  represents length.

59. (c) Since,  $T^2 = \frac{4\pi^2 r^3}{GM}$

$$[T^2] = \frac{[L]^3}{[M^{-1} L^3 T^{-2}][M]}$$

(Dimension of  $G$  is  $[M^{-1} L^3 T^{-2}]$ )

$$[T^2] = \frac{[L]^3}{[L^3 T^{-2}]} = [T^2]$$

60. (b) Dimension formula for  $[B] = L^2$

$$A = \frac{x^2}{tE} = \frac{L^2}{T^2 \cdot T^{-2} \cdot T^{-1}} = \frac{1}{MT^{-1}}$$

Dimension formula for  $[A] = M^{-1} T$

Dimension formula for  $[AB]$

$$= L^2 M^{-1} T^2$$

61. (a) Force,  $F = ax^2 + bt^2$

$$[a] = \left[ \frac{F}{x^2} \right] = [M^1 L^{-1} T^{-2}]$$

$$[b] = \left[ \frac{F}{t^2} \right] = [M^1 L^{-1} T^{-4}]$$

$$\left[ \frac{b^2}{a} \right] = \frac{[M^2 L^{-2} T^{-8}]}{[M^1 L^{-1} T^{-2}]} = [M^1 L^{-1} T^{-6}]$$

62. (a)  $m = k C^{1/2} h^{1/2}$

$$M^1 L^1 T^0 = [L^{-1/2} P^{1/2}] [M^{-1/2} L^{1/2} T^{-1/2}]^2 [MLT^{-1}]^2$$

By comparing  $P = 1/2$

63. (b)  $[P] = \left[ \frac{a}{V^2} \right] = [a] = [PV^2]$

And dimension of  $V$  = dimension of  $b$

$$\therefore \left[ \frac{a}{b^2} \right] = \left[ \frac{[PV^2]}{[V]^2} \right] = [P]$$

64. (d) According to the question,  $m \propto k r C$

$$M^1 L^1 T^0 = (MLT^{-2})^k (LT^{-1})^k (M^1 L^3 T^{-2})^k$$

$$M^1 L^1 T^0 = M^{k+3k} L^{k+3k} T^{-2k-2k}$$

$$\text{on comparing both sides}$$

$$x - z = 1 \quad \dots (i)$$

$$2x + y + 3z = 0 \quad \dots (ii)$$

$$-x - y - 2z = 0 \quad \dots (iii)$$

On solving equations (i), (ii) and (iii) we get

$$x = \frac{1}{2}, y = \frac{1}{2}, z = \frac{-1}{2}$$

65. (c)  $b$  has the dimensions of volume,  $[b] = [V]$

$$l \left[ \frac{a}{V^2} \right] = [P]$$

$$\Rightarrow \left[ \frac{a}{b^2} \right] = [P] \therefore \left[ \frac{b^2}{a} \right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

$$\therefore \left[ \frac{b^2}{a} \right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

66. (b)  $(x - At)^2 + \left( y - \frac{1}{B} \right)^2 = a^2$

$$[At] = [L]$$

$\therefore [A] = [T^{-1}]$  {dimension of  $A$ }  
Similarly, given

$$\left[ \frac{t}{B} \right] = [L] = \frac{1}{T[B]} = [L]$$

$$\therefore [B] = [T^{-1} L^{-1}]$$

67. (a)  $[T^{-1}] = [L]^a [M^1 L^3 T^{-2}]^b \left[ \frac{M}{L} \right]^c$

$$\Rightarrow [T^{-1}] = [M^{b+c} L^{-2b-1} T^{-2b}]$$

Equating the power,

$$c = \frac{1}{2}, b = -\frac{1}{2}, a - 3b = 0 \Rightarrow a = \frac{1}{2}$$

68. (b)  $[ML^{-2}] = [MLT^{-2}]^a [LT^{-1}]^b [M]^{1/2}$

$$= [M^{1+a+b/2} L^{-2a-b} T^{-2a-b}]$$

$$a = 1,$$

$$a + b = -3,$$

$$2a - b + c = 0$$

$$\text{Solving (i), (ii) and (iii)}$$

$$c = -2 \Rightarrow b = -4$$

69. (a)  $X$  and  $\frac{a}{y^2}$  have same dimension

$$Y \text{ and } b \text{ have same dimensions}$$

$$\therefore [a] = [ML^2 T^{-2}] [b] = [L]$$

$$\left[ \frac{a}{b} \right] = [ML^2 T^{-2}] \text{ has dimension}$$

energy

70. (a) Given that,

$$v_2 = \frac{n}{m^2} v_1 \Rightarrow \frac{L}{T} = \frac{n}{m^2} \frac{L}{T}$$

$$\text{Also, } a_1 = a/mn \Rightarrow \frac{L}{T^2} = \frac{L}{T^2} \frac{1}{m}$$

$$\Rightarrow \left( \frac{T_1}{T_2} \right)^2 = \frac{L_1}{L_2} \times \frac{1}{mn}$$

$$\Rightarrow \frac{T_1^2}{T_2^2} = \frac{T_1}{T_2} \times \frac{m^2}{n} \times \frac{1}{mn}$$

$$\Rightarrow \frac{T_1}{T_2} = T_2$$

$$\text{Solving (i) and (ii), we get}$$

$$L_1 = \frac{n^2}{m^2} L_2$$

71. (d) Given,  $u = \frac{\alpha}{\beta} \sin \left( \frac{\alpha x}{\beta} \right)$

Dimension of Boltzmann Constant

$$[k] \rightarrow [ML^2 T^{-2} K^{-1}]$$

Dimension of Energy density

$$[u] \rightarrow [ML^{-1} T^{-2}]$$

Dimension of Temperature  $[T] = [K]$

Since, angle has no dimension, therefore its dimensional formula is  $[M^0 L^0 T^0]$

$$\text{or } \frac{\alpha x}{kt} = \left[ \frac{MLT^{-2}}{MLT^{-2}} \right]$$

$$\Rightarrow \frac{\alpha [L]}{[ML^2 T^{-2} K^{-1} K]} = [M^0 L^0 T^0]$$

$$\alpha = [MLT^{-2}]$$

$$\text{Also, dimension of } \frac{\alpha}{\beta} = \text{dimension of energy density}$$

$$\frac{\alpha}{\beta} = [ML^{-1} T^{-2}]$$

$$\beta = \frac{[MLT^{-2}]}{[ML^{-1} T^{-2}]} = [M^2 L^2 T^{-2}]$$

72. (d) Trigonometric function and logarithmic function are dimensionless quantities.

$$\therefore [\eta] = [M^0 L^0 T^0]$$

Also, dimensions of temperature,

$$[T] = [M^0 L^0 T^0 K]$$

Dimensions of Boltzmann constant,  $[k]$

$$= [MLT^{-2} K^{-1}]$$

$$\text{Dimension of } [x] = [M^0 L^0 T^0]$$

$$A. [B] = \left[ \frac{kt}{x} \right] = \left[ \frac{E}{x} \right] = [MLT^{-2}] = [F]$$

$$B. [a\beta] = [M^2 L^2 T^{-2}] \Rightarrow [a]^{-1} = [B] = \left[ \frac{kt}{x} \right]$$

$$\text{So } [a]^{-1} [x] = [kt] = [MLT^{-2}]$$

$$C. \eta \sin \theta = a\beta$$

$$\text{So, } [\eta \sin \theta] = [a\beta]$$

$[\eta] = [M^0 L^0 T^0 K]$  it is dimensionless quantity

$$D. [a] \neq [B]$$

73. (c)  $\frac{kt}{\beta x} = 1 \Rightarrow [B] = \left[ \frac{MLT^{-2}}{L} \right]$

$$\left( \because E = \frac{1}{2} kt \right)$$

As  $P$  is dimensionless

$$\Rightarrow [a] = [B] = [MLT^{-2}]$$

74. (c) From principle of homogeneity,

$$[P] = \left[ \frac{a}{V^2} \right] \text{ and } [b] = [V] = \left[ \frac{a}{b} \right] = [PV]$$

75. (c) Let us suppose,

$$\text{Mass } \alpha [r^a v^b k^c] \Rightarrow [M^1 L^3 T^0] = [T^a] \times [L^{3a-b-c}]$$

On equating both sides, we get

$$\Rightarrow a - b - c = 0; c = 1, \text{ and } b + 2c = 0$$

$$\Rightarrow b = -2c = -2$$

$$\Rightarrow a = b + c = 1 - 2 = -1$$

$$\text{Hence, dimension of mass} = [r^a v^b k^c] = [r^{-1} v^{-2} k^1]$$

76. (b) We have,  $W = \alpha^2 \beta e^{\frac{h\nu}{kT}}$

$$\therefore [\beta x^2] = [kt] = [ML^2 T^{-2}] \left[ \because E = \frac{1}{2} kt \right]$$

$$\Rightarrow [\beta] = [MT^{-2}]$$

$$\text{And, } [W] = [\alpha^2 \beta] \Rightarrow [ML^2 T^{-2}] = [\alpha^2] [MT^{-2}]$$

$$\Rightarrow [\alpha] = [L]$$

77. (c) Work done,

$$[W] = [F] [d] = [MLT^{-2}] \rightarrow \text{correct}$$

$$\text{height, } [h] = [L] = \left[ \frac{2s \cos \theta}{\rho g} \right] \rightarrow \text{correct}$$

$$\text{Volume, } [V] = [L]^3; \left[ \frac{\pi \rho a^2}{8 \eta L} \right] = [L^3 T^{-1}] \rightarrow \text{incorrect}$$

$$\text{Current density, } J = e \frac{\partial E}{\partial t} \rightarrow \text{correct}$$

78. (c)  $\frac{1}{4\pi\epsilon_0} \frac{q_1^2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_2^2}{r^2} \times \frac{r^2}{\lambda \times \lambda} = \frac{F \times r}{E \times \lambda}$

Dimension of 'Fr' (work) and 'E' (energy) are same.

Therefore, given quantity is dimensionless

79. (a) Since,  $\frac{\alpha^2}{\beta kT}$  should be dimensionless.

$$\therefore [\alpha] = \frac{L^2}{ML^2 T^{-2}} = M^{-1} T^2$$

$$[KT] = [\text{Energy}]$$

Dimension of  $a\beta^2$  should be dimension of  $W$ .

$$\text{So, } [a\beta^2] = MLT^{-2}$$

$$\Rightarrow [\beta^2] = \frac{ML^2 T^{-2}}{M^2 T^2} = M^2 L^2 T^{-4}$$

$$\Rightarrow [\beta] = MLT^{-2}$$

80. (b) Let  $M = K (V)^a (F)^b (T)^c$

$$\Rightarrow \text{Equating dimensions of both the sides}$$

$$M^1 L^1 T^0 = ([L^3 T^{-1}]^a [ML^{-1} T^{-2}]^b [T]^c)$$

$$MLT^0 = M^{b-a} L^{3a-b} T^{-2b+c}$$

$$\text{Comparing dimensions, we get}$$

$$a = -1, b = 1, c = 1$$

$$\text{Thus, } M = (\text{Some Number}) (V^{-1} F^1 T^1)$$

$$\Rightarrow [M] = [V^{-1} F^1 T^1]$$

81. (c)  $[A] = [MLT^{-2}]$

$$[B] = [L^{-1}]$$

$$[D] = [T^{-1}]$$

$$\Rightarrow \left[ \frac{AD}{B} \right] = \frac{[MLT^{-2}][T^{-1}]}{[L^{-1}]} \text{ or}$$

$$\left[ \frac{AD}{B} \right] = [MLT^{-3}]$$

82. (\*)  $V = K(h)^a (Y)^b (G)^c$  ( $V$  is voltage)

$$\text{We know, } [h] = ML^2 T^{-1}$$

$$[I] = A, [G] = M^1 L^1 T^{-2}$$

$$[C] = L T^{-1}, [V] = ML^2 T^{-1} A^{-1}$$

$$ML^2 T^{-1} A^{-1} = (ML^2 T^{-1})^a (A)^b (ML^1 T^{-2})^c$$

$$(LT^{-1})^d$$

$$ML^2 T^{-1} A^{-1} = M^{a+b+c} L^{2a+b+d} T^{-a-2c-d} A^{-b}$$

$$x - z = 1 \quad \dots (i)$$

$$2x + 3z + w = 2 \quad \dots (ii)$$

$$-x - 2z - w = -3 \quad \dots (iii)$$

$$y = -1 \quad \dots (iv)$$

On solving,

$$z = -1, x = 0, w = 5, y = -1$$

Hence,  $V = K(h)^0 (Y)^{-1} (G)^{-1} (C)^1$  (Bonus)

83. (a)  $K_B = \frac{QDr}{ADT}$

$$\text{Dimensional formula} = \frac{[ML^2 T^{-2}][L]}{[L^2][K][T]}$$

$$= M^1 L^{-1} T^{-1} K^{-1}$$

84. (d) Solar constant =  $\frac{E}{AT}$

$$\frac{M^1 L^2 T^{-2}}{L^2 T} = M^1 T^{-3}$$

85. (c)  $x = \frac{IFv^2}{WL^4}$

$$\therefore [x] = \frac{[ML^2] \times [MLT^{-2}] \times [LT^{-1}]^2}{[ML^2 T^{-2}] \times L^4}$$

$$= ML^{-1} T^{-2} = [\text{Energy density}]$$

86. (a) Planck's constant,  $(h) = [ML^2 T^{-1}]$

$$\text{Speed of light, } (c) = [LT^{-1}]$$

$$\text{Universal gravitational constant}$$

$$(G) = [M^{-1} L^3 T^{-2}]$$

$$[f] = \frac{\sqrt{M^1 L^1 T^{-1}} \times L \times T^{-1}}{\sqrt{M^1 L^1 T^{-1}}} = M^1 L^2 T^{-2}$$

87. (a) [Young's modulus] =  $\left[ \frac{\text{Force}}{\text{Area}} \right]$

91. (b) Power of exponential should be dimensionless.

$$\begin{aligned} [L^{-1}] &= [a]b \Rightarrow [L] = [a] [ML^2 T^{-2}] \\ &\Rightarrow [a] = [M^{-1} T^2] \\ [F] &= [a][B] \Rightarrow [MLT^{-2}] = [M^{-1} T^2][B] \\ &\Rightarrow [B] = [N^2 L^{-2}] \end{aligned}$$

92. (c) (i)  $[F] = [F]^2 [Y]^2 [A]^2$

$$[ML^{-2} T^{-2}] = [MLT^{-2}]^2 [LT^{-1}]^2 [L^2 T^{-2}]$$

$$[ML^{-2} T^{-2}] = [M^2 L^4 T^{-6}]$$

$$\therefore a = 1 \quad \dots (i)$$

$$a - b + c = -1 \quad \dots (ii)$$

$$-2a - b - 2c = -2 \quad \dots (iii)$$

After solving above equations (i), (ii) and (iii).

$$a = 1, b = -4, c = 2 \therefore [Y] = [P^{-4} A^2 F]$$

93. (b) Time constant  $\tau = RC \Rightarrow [RC] = T$

$$\Rightarrow R = \frac{T}{C} \Rightarrow \left[ \frac{L}{RCV} \right] = \left[ \frac{L}{T} \right] = T^{-1}$$

$$\text{Thus, } \left[ \frac{L}{RCV} \right] = T^{-1} = A^{-1}$$

94. [15]  $\left[ \frac{dC}{C} \right] = \frac{dP}{P} + \frac{2dq}{q} + \frac{3dr}{r} + \frac{1ds}{s}$

$$= 15\%$$

95. (a)  $\rho = \frac{M}{\pi r^2 l}$

The maximum percentage error in the measurement of the density of a wire

$$\frac{\Delta \rho}{\rho} = \frac{\Delta M}{M} + \frac{\Delta r}{r} + \frac{\Delta l}{l}$$

$$= \frac{\Delta \rho}{\rho} = \left( \frac{0.003}{0.6} + \frac{2 \times 0.01}{0.5} + \frac{0.05}{10} \right) \times 100$$

96. (c)  $E(t) = \alpha e^{\beta t}$

$$E(t) = 3\alpha e^{\beta t} \alpha \beta da - \beta \alpha e^{\beta t} \beta dt$$

$$\frac{dE(t)}{E(t)} = \frac{3\alpha e^{\beta t} \alpha \beta da - \beta \alpha e^{\beta t} \beta dt}{E(t)}$$

$$\frac{\Delta E}{E} = \frac{3\Delta \alpha}{\alpha} + \beta \Delta t$$

$$\frac{\Delta E}{E} \times 100 = 1.6 \quad (E = 5.8) \Rightarrow \Delta E = 0.08$$

$$\frac{\Delta E}{E} = 3 \times 0.012 + 0.3 \times 0.08$$

$$\Rightarrow \frac{\Delta E}{E} = 0.06$$

$$\Rightarrow \frac{\Delta E}{E} = 6\%$$

$$\text{maximum percentage error in the energy is: } \frac{\Delta E}{E} = 6\%$$

97. (b)  $y = \frac{32.3 \times 1125}{27.4}$

$$y = 1326.186$$

Significant Figures Rule:

The given numbers have 3 significant digits.

The result must also be rounded to 3 significant digits.

Rounding 1326.186 to 3 significant digits:

1326.186  $\rightarrow$  1330 (since 1326 rounds up to 1330)

the correct answer is (b) 1330.

98. [3] Least Count =  $\frac{0.75}{15} = 0.05$

$$A = lb$$

$$\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b}$$

$$= \frac{0.05}{5} + \frac{0.05}{2.5} = \frac{1}{100} + \frac{1}{50} = \frac{x}{100} = \frac{3}{100}$$

$$x = 3$$

99. (c) For all medium

(a) can also be correct in Hindi & Gujarati medium (NTA)

$$B = \frac{\mu_0 m}{4\pi r^2}$$

$$\frac{\Delta B}{B} = 3 \times \left( \frac{\Delta r}{r} \right)$$

$$\% \text{ uncertainty in } B = 3\%$$

If we also consider error in  $\ell$ .

$$\frac{\Delta B}{B} = \frac{\Delta \ell}{\ell} + 3 \times \left( \frac{\Delta r}{r} \right) = 1 + 3 \times 1 = 4\%$$

100. [7000]  $Q = \frac{ab^2}{cd}$

$$\Rightarrow \frac{\Delta Q}{Q} = \frac{\Delta a}{a} + 4 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d}$$

$$\text{Percentage error in } Q = \left[ \frac{\Delta a}{a} + 4 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d} \right] \times 100$$

$$= \left[ \frac{3}{60} + 4 \left( \frac{0.1}{20} \right) + \left( \frac{0.2}{40} \right) + \left( \frac{0.1}{50} \right) \right] \times 100 = \frac{x}{1000}$$

$$\Rightarrow x = 7700$$

101. (c) For finding percentage error in  $Y$ ,

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta \ell}{\ell}$$

$$= \frac{5}{500} + \frac{0.02}{2} = 0.01 + 0.01$$

$$\frac{\Delta Y}{Y} = 0.02 \Rightarrow \% \frac{\Delta Y}{Y} = 2\%$$

102. (b) if  $a \leq 5$  order is  $b$

$a > 5$  order is  $b + 1$

103. (d) From  $T = 2\pi \sqrt{\frac{m}{k}}$

$$\text{Since, } T^2 \propto \frac{m}{k}$$

Therefore,

$$\frac{\Delta T^2}{T^2} = \frac{\Delta m}{m} - \frac{\Delta k}{k}$$

$$\frac{\Delta k}{k} = \frac{\Delta m}{m} - \frac{\Delta T^2}{T^2}$$

$$\frac{\Delta k}{k} = (-1\%) - 2(2\%) = 5\% = 5\%$$

104. (c) Sum of number by considering significant digit sum =  $4.6 + 4.6 + 4.6$ ,  $4.6 = 18.4$

$$\text{Arithmetic Mean} = \frac{\text{sum}}{4} = \frac{18.4}{4} = 4.6$$

105. (b) According to lens

$$\text{Formula,}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow f^{-1} = v^{-1} - u^{-1}$$

$$\text{differentiating w.r.t } f$$

$$\Rightarrow -f^{-2} df = -v^{-2} dv - u^{-2} du$$

$$\Rightarrow df = \frac{dv}{v^2} + \frac{du}{u^2}$$

$$\Rightarrow df = f^2 \left[ \frac{dv}{v^2} + \frac{du}{u^2} \right]$$

106. (c) Conceptual

107. (b) Resistivity of the given material,

$$\rho = R \frac{l}{A}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + 2 \frac{\Delta r}{r} + \frac{\Delta \ell}{\ell}$$

$$= \frac{10}{100} + 2 \times \frac{0.05}{0.35} + \frac{0.2}{15}$$

$$= \frac{1}{10} + \frac{2}{7} + \frac{1}{75} = 39.9\%$$

108. (c) Time period of pendulum,

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\text{Percentage error, } \frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$= \frac{0.2}{20} + 2 \left( \frac{1}{40} \right) = \frac{0.3}{20}$$

$$\text{Percentage change} = \frac{0.3}{20} \times 100 = 6\%$$

109. (b) Resistance,  $R = \frac{\rho l}{A}$

$$= \left( \frac{2}{a} \Delta a + \frac{3}{b} \Delta b + \frac{\Delta c}{c} + \frac{1}{2} \frac{\Delta d}{d} \right) \times 100\%$$

$$= 2(1\%) + 3(2\%) + 3\% + \frac{1}{2} \times 4\%$$

$$= 2\% + 6\% + 3\% + 2\% = 13\%$$

114. (d) Terminal velocity of a spherical body in liquid

$$V_T = \frac{2}{9} r^2 \frac{(\rho - \sigma)g}{\eta} \Rightarrow V_T \propto r^2$$

$$\Rightarrow \frac{\Delta V_T}{V_T} = 2 \frac{\Delta r}{r}$$

110. (a) Given  $V = (200 \pm 5) \text{ V}$   
 $I = (20 \pm 0.2) \text{ A}$

$$R = \frac{V}{I}$$

$$\text{According to error analysis}$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} = \frac{5}{200} + \frac{0.2}{20} = \frac{5}{200} + \frac{2}{200}$$

$$\frac{\Delta R}{R} = \frac{7}{200}$$

$$\% \text{ error } \frac{\Delta R}{R} \times 100 = \frac{7}{200} \times 100 = 3.5\%$$

111. (c)  $Q = \frac{a^2 b^3}{c^2}$

By using the percentage error formula,

$$\frac{\Delta Q}{Q} = 4 \frac{\Delta a}{a} + 3 \left( \frac{\Delta b}{b} \right) + 2 \frac{\Delta c}{c}$$

$$\left( \frac{\Delta Q}{Q} \times 100 \right) = 4 \left( \frac{\Delta a}{a} \times 100 \right) + 3 \left( \frac{\Delta b}{b} \times 100 \right) + 2 \left( \frac{\Delta c}{c} \times 100 \right)$$

$$\% \text{ error in } Q = 4 \times 3\% + 3 \times 4\% + 2 \times 5\%$$

$$= 12\% + 12\% + 10\% = 34\%$$

112. (c) Least count is 0.2 cm

$$u = (100 \pm 0.2) - (80 \pm 0.2) = (20 \pm 0.4) \text{ cm}$$

$$v = (180 \pm 0.2) - (100 \pm 0.2) = (80 \pm 0.4) \text{ cm}$$

From lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{80} - \frac{1}{20} = \frac{-20 - 80}{-1600}$$

$$f = 16 \text{ cm}$$

$$\text{Also } \frac{\Delta f}{f} = \frac{\Delta v}{v} + \frac{\Delta u}{u}$$

$$\therefore \frac{\Delta f}{f} \times 100 = \left( \frac{\Delta v}{v} + \frac{\Delta u}{u} \right) \times 100$$

$$\Rightarrow \% f = \left( \frac{0.4}{80} + \frac{0.4}{20} \right) \times 100 = 1.70$$

113. (a) Percentage error in quantity  $P$ ,

$$\frac{\Delta P}{P} \times 100\%$$

$$= \left( \frac{2}{a} \Delta a + \frac{3}{b} \Delta b + \frac{\Delta c}{c} + \frac{1}{2} \frac{\Delta d}{d} \right) \times 100\%$$

$$= 2(1\%) + 3(2\%) + 3\% + \frac{1}{2} \times 4\%$$

$$= 2\% + 6\% + 3\% + 2\% = 13\%$$

114. (d) Terminal velocity of a spherical body in liquid

$$V_T = \frac{2}{9} r^2 \frac{(\rho - \sigma)g}{\eta} \Rightarrow V_T \propto r^2$$

$$\Rightarrow \frac{\Delta V_T}{V_T} = 2 \frac{\Delta r}{r}$$

$$\Rightarrow \frac{\Delta V_T}{V_T} \times 100\% = 2 \left( \frac{0.1}{5} \right) \times 100 = 4\%$$

115. (c) When  $R_1$  and  $R_2$  are connected in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \quad \dots (i)$$

$$\left[ R = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 15}{10 + 15} = 6 \right]$$

Differentiating (i) both sides

$$\frac{\Delta R}{R} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2}$$

$$\Rightarrow \frac{\Delta R}{R} = \left( \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} \right) R$$

$$= \left( \frac{0.5}{100} + \frac{0.5}{225} \right) 6 = \frac{13}{300}$$

$$\frac{\Delta R}{R} \times 100 = \frac{13}{3} = 4.33\%$$

116. (a) K.E.  $= K \pm \Delta K$

$$= K \pm K \left( \frac{\Delta m}{m} + \frac{2\Delta v}{v} \right) \therefore \frac{\Delta K}{K} = \frac{\Delta m}{m} + \frac{2\Delta v}{v}$$

$$= 1000 \pm 1000 \left( \frac{0.5}{5} + \frac{2 \times 0.4}{20} \right)$$

$$= (1000 \pm 140) \text{ J}$$

117. (c)  $\rho = \frac{m}{\pi r^2 l} \Rightarrow \left[ \frac{\Delta \rho}{\rho} \right] = \left[ \frac{\Delta m}{m} \right] + 2 \left[ \frac{\Delta r}{r} \right] + \left[ \frac{\Delta l}{l} \right]$

$$= \frac{0.01}{0.4} + \frac{2(0.03)}{6} + \frac{0.04}{8} = 0.04$$

$$\text{Percentage error in density}$$

$$= \left( \frac{\Delta \rho}{\rho} \right) \times 100\% = 4\%$$

118. (c)  $z = \frac{a^2 b^3}{c^4}$

The relative error in  $z$  can be given as

$$\frac{\Delta z}{z} = \frac{2\Delta a}{a} + \frac{3\Delta b}{b} + \frac{4\Delta c}{c}$$

$$= \frac{2}{4} + \frac{3}{4} + \frac{4}{4} = \frac{9}{4} = 2.25$$

119. [1.99]  $L = 1 \text{ m}$ ,  $\Delta L = 0.4 \times 10^{-3} \text{ m}$

$$m = 1 \text{ kg}$$
,  $d = 0.4 \times 10^{-3} \text{ m}$

$$\text{Area, } A = \frac{\pi d^2}{4} = \frac{3.14 \times (0.4 \times 10^{-3})^2}{4}$$

$$= 0.1256 \times 10^{-6} \text{ m}^2$$

$$\text{Young's modulus}$$

$$Y = \frac{FL}{A\Delta L} = \frac{(1 \times 10^3) \times 1}{0.1256 \times 10^{-6} \times 0.4 \times 10^{-3}}$$

$$= 1.99 \times 10^{12} \text{ N/m}^2$$

$$\text{Now, } \frac{\Delta Y}{Y} = \frac{\Delta F}{F} + \frac{\Delta L}{L} + \frac{\Delta A}{A} + \left( \frac{\Delta \Delta L}{\Delta L} \right)$$

$$= \frac{2\Delta d}{d} + \frac{\Delta d}{d} = 2 \times \frac{0.01}{0.4} + \frac{0.02}{0.4}$$

$$\Delta Y = 0.1 \times 1.99 \times 10^{12} = 1.99 \times 10^{11} = 1.99$$

120. (b) For torque

$$[\tau] = [\text{MLT}^2]$$

So, percentage error in torque  $\% \tau$

$$= \% M + 2\% L + 2\% T = 25\%$$

121. (a) Given,  $m = 0.6 \pm 0.006$ ,  $r = 0.5 \pm 0.005$ ,  $I = 4 \pm 0.04$

Percentage error in  $m$ ,

$$\frac{\Delta m}{m} \times 100 = \frac{0.006}{0.6} \times 100 = 1\%$$

Percentage error in  $r$ ,

$$\frac{\Delta r}{r} \times 100 = \frac{0.005}{0.5} \times 100 = 1\%$$

Percentage error in  $I$ ,

$$\frac{\Delta I}{I} \times 100 = \frac{0.04}{4} \times 100 = 1\%$$

Formula of density  $\rho = \frac{m}{V} = \frac{m}{\pi r^2 l}$

$$\therefore \frac{\Delta \rho}{\rho} \times 100 = \frac{\Delta m}{m} \times 100 + 2 \frac{\Delta r}{r} \times 100 + \frac{\Delta I}{I} \times 100$$

$$= 1 + 2 \times 1 + 1 = 4\%$$

122. [18] We have,  $z = x^2 y^2$

$$\frac{\Delta z}{z} \times 100 = 3 \frac{\Delta x}{x} \times 100 + 2 \frac{\Delta y}{y} \times 100$$

$$= \left( 3 \times 4 + \frac{1}{2} \times 12 \right) \times 100 = 18\%$$

123. [150] It is given that  $x_1 = 1.22 \text{ mm}$ ,  $x_2 = 1.23 \text{ mm}$

$$x_1 = 1.19 \text{ mm and } x_2 = 1.20 \text{ mm}$$

$$x_{\text{mean}} = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$$= \frac{1.22 + 1.23 + 1.19 + 1.20}{4} = 1.21$$

(Measured data are up to two digits after decimal)

$$\therefore d = (5.54 \pm 0.07) \text{ mm}$$

126. (c)

$$\frac{\Delta T}{T} = \left( \frac{\Delta m}{m} \right) + \left( \frac{\Delta g}{g} \right) + \left( \frac{\Delta L}{L} \right) + \left( \frac{\Delta T}{T} \right) + \left( \frac{\Delta T}{T} \right)$$

$$= \left( \frac{1\text{g}}{1\text{kg}} \right) + 0 + 2 \left( \frac{\Delta T}{T} \right) + \left( \frac{\Delta L}{L} \right) + \left( \frac{\Delta T}{T} \right)$$

$$= \left( \frac{1\text{g}}{1\text{kg}} \right) + 2 \left( \frac{0.001\text{cm}}{0.2\text{cm}} \right) + \left( \frac{0.001\text{cm}}{0.5\text{cm}} \right)$$

$$+ \left( \frac{0.001\text{m}}{1\text{m}} \right)$$

$$= \frac{1+10+2+1}{1000} \times 100\% = 1.4\%$$

127. (b) Here, time period of oscillation,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\therefore T^2 = 4\pi^2 \left( \frac{L}{g} \right) \Rightarrow g = 4\pi^2 \left( \frac{L}{T^2} \right)$$

$$\therefore \frac{\Delta g}{g} = \left( \frac{\Delta L}{L} + \frac{2\Delta T}{T} \right) \times 100\%$$

$$= \left[ \frac{1\text{mm}}{1\text{m}} + \frac{2 \times 0.01}{1.95} \right] \times 100\%$$

$$= (0.001 + 0.0102) \times 100 = 1.13\%$$

128. (a)  $V = IR = I \cdot \frac{\rho L}{A} \Rightarrow \rho = \frac{\pi d^2 V}{4 I L}$

$$\text{So, } \frac{\Delta \rho}{\rho} = \frac{\Delta d}{d} + \frac{\Delta V}{V} + \frac{\Delta L}{L} + \frac{\Delta I}{I}$$

$$= 2 \times \frac{0.01}{5.00} + \frac{0.1}{5.0} + \frac{0.1}{10.0} + \frac{0.01}{2000} = 0.039$$

$$\text{Percentage error} = \frac{\Delta \rho}{\rho} \times 100 = 3.9\%$$

129. (a) Initial resistance,  $R_i = \frac{\rho L}{A}$ ,  $\ell_i = 1.25 \ell$

$$\text{As volume of wire will remain same,}$$

$$A_i = \frac{A}{1.25}$$

$$\text{Final resistance,}$$

$$R_f = \frac{\rho (1.25\ell)}{(A/1.25)} = (1.25)^2 \times \frac{\rho \ell}{A}$$

$$R_f = 1.5625 R_i$$

$$\% \text{ change in resistance}$$

$$= \frac{R_f - R_i}{R_i} \times 100 = 56.25\%$$

130. [1] Time period of pendulum

$$T = 2\pi \sqrt{\frac{\ell}{g}} \therefore g = 4\pi^2 \frac{\ell}{T^2}$$

$$\text{Percentage error in } g,$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

(Where  $\Delta T_i$  is least count for time and  $h$  is number of complete oscillation) From the given table it is clear that student - 1 will get least error.

131. (b)  $T = 2\pi \sqrt{\frac{L}{g}}$

$$\Rightarrow \frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta L}{L} \times 100 + \frac{1}{2} \frac{\Delta g}{g} \times 100$$

$$[\text{As } \Delta g = 0]$$

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2} \times 0.1 = 0.05\%$$

$$\text{For 1 day } \Delta T = \frac{0.05}{100} \times 24 \times 3600$$

$$\Delta T = 43.2 \text{ sec}$$

132. (b)  $T = 2\pi \sqrt{\frac{L}{g}} \Rightarrow T^2 = 4\pi^2 \left( \frac{L}{g} \right) \Rightarrow g = \frac{4\pi^2 L}{T^2}$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{1 \times 10^{-3}}{1 \times 10^{-2}} + \frac{2 \times 1}{100}$$

$$\Rightarrow \frac{\Delta g}{g} = 0.02 + 0.01 = 0.03$$

$$\Rightarrow 100 \times \frac{\Delta g}{g} = 0.03 \times 100 = 3\%$$

133. (b)

$$\left( \frac{\Delta x \times 100}{x} \right)$$

$$= \left[ 2 \times \frac{\Delta m}{m} + \frac{4 \times \Delta r}{r} + \frac{x \times \Delta g}{g} + \frac{3}{2} \frac{\Delta \ell}{\ell} \right] \times 100$$

$$\Rightarrow 18 = (2 \times 1 + (4 \times 0.5) + (x \times 1) + \left( \frac{3}{2} \times 4 \right)) \Rightarrow 8 = x \times p$$

$$\text{Using given options}$$

$$x = \frac{16}{2} \therefore p = \frac{3}{2}$$

134. (b)  $\gamma = \frac{4\pi G \rho^2}{3b^3}$

$$\frac{\Delta \gamma}{\gamma} = \frac{\Delta M}{M} + \frac{3\Delta L}{L} + \frac{\Delta b}{b} + 3 \frac{\Delta \rho}{\rho}$$

$$= \frac{1 \times 10^{-3}}{2} + \frac{3 \times 10^{-1}}{4} + \frac{10^{-2}}{4} + \frac{0.01 \times 10^{-1}}{0.4} + \frac{10^{-2}}{5}$$

$$= 10^{-1} \left[ \frac{1}{2} + \frac{3}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{5} \right] = 0.0155$$

135. [14]  $T = 2\pi \sqrt{\frac{I}{g}} \Rightarrow I = \frac{T^2 g}{4\pi^2}$

$$E = mgl \frac{\theta^2}{2} = mg \frac{T^2 \theta^2}{8\pi^2}$$

$$\Rightarrow \frac{\Delta E}{E} = 2 \left( \frac{\Delta g}{g} + \frac{\Delta T}{T} \right) = 2(4 + 3) = 14\%$$

$$136. [5] \epsilon_1 = \frac{\Delta s}{s} \times 100 = 4\%, \epsilon_2 = \frac{\Delta t}{t} \times 100 = 1\%$$

$$\therefore \epsilon_2 = \pm (\epsilon_1 + \epsilon_2) = \pm (4\% + 1\%) = \pm 5\%$$

137. [34] We know that, volume of sphere,

$$V = \frac{4\pi}{3} r^3 \Rightarrow \frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\% \text{ error in volume} = \frac{\Delta V}{V} \times 100 = 3 \frac{\Delta r}{r} \times 100$$

$$= 3 \times \frac{0.85}{7.50} \times 100 = 34$$

$$138. (a) \frac{\Delta T}{T} = \frac{1}{2} \left( \frac{\Delta g}{g} + \frac{\Delta L}{L} \right)$$

$$\frac{\Delta g}{g} = \frac{2\Delta T}{T} + \frac{\Delta L}{L} = 2 \left( \frac{1}{50} \right) + \frac{0.1}{25.0} = 4\%$$

$$139. (b) \frac{\Delta z}{z} = \frac{2}{3} \frac{\Delta a}{a} + \frac{2}{3} \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c} + \frac{3}{2} \frac{\Delta d}{d}$$

$$= 2 \times 2 + \frac{2}{3} \times 1.5 + \frac{1}{2} \times 4 + 3 \times 2.5$$

$$14.5\%$$

$$140. (c) \rho = \frac{m}{\frac{4}{3} \pi \left( \frac{d}{2} \right)^3}$$

$$\therefore \% \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 3 \left( \frac{\Delta d}{d} \right)$$

$$= 6 + 3 \times 1.5 = 10.5\%$$

141. (\*) As the error is larger in length measurement. So to find the error in density measurement.

$$\rho_{\text{min}} = \frac{m_{\text{min}}}{V_{\text{min}}} = \frac{9.9}{(0.11)} = 7438 \text{ kg/m}^3$$

$$\rho_{\text{max}} = \frac{m_{\text{max}}}{V_{\text{min}}} = \frac{10.1}{(0.09)} = 13854.6 \text{ kg/m}^3$$

$$\therefore \Delta \rho = 13854.6 - 7438 = 6416.6 \text{ kg/m}^3$$

$$(10.1)$$

$$142. (c) \text{ In case of simple pendulum,}$$

$$T = 2\pi \sqrt{\frac{L}{g}} \therefore g = \frac{4\pi^2 L}{T^2}$$

$$\% \text{ error in } g$$

$$= \frac{\Delta g}{g} \times 100 = \left( \frac{\Delta L}{L} + \frac{2\Delta T}{T} \right) 100\%$$

$$= \left( \frac{0.1}{55} + \frac{2 \left( \frac{1}{20} \right)}{\frac{30}{20}} \right) 100\% = 6.8\%$$

$$143. (c) V = \pi R^2 h = \frac{\pi}{4} D^2 h = 4260 \text{ cm}^3$$

$$\therefore \frac{\Delta V}{V} = 2 \frac{\Delta D}{D} + \frac{\Delta h}{h}$$

$$\Delta V = \left( 2 \times \frac{0.1}{12.6} + \frac{0.1}{34.2} \right) V$$

$$= \frac{2 \times 426}{12.6} + \frac{426}{34.2}$$

$$= 67.61 + 12.459 = 80.075$$

$$\therefore V = 4260 \pm 80 \text{ cm}^3$$

144. (c)  $R_s = (1.005)^2 R$

$$\% \epsilon_s = \frac{R_s - R}{R} \times 100 = 1\%$$

145. (b) A vernier scale is designed such that  $n$  divisions are slightly smaller than the main scale divisions. This difference allows precise measurements by aligning a particular vernier division with the main scale.

The vernier constant (least count) is actually given by:

Least Count = Value of One Main Scale Division - Value of One Vernier Scale Division

Thus, the correct option is:

A is true but B is false.

146. [35] The least count is given by the formula:

$$LC = \frac{\text{Pitch}}{\text{Number of circular divisions}} = \frac{p}{N}$$

$$= 0.01 \text{ mm}$$

Increase in Pitch: The pitch is increased by 75%. So, the new pitch becomes:

$$\text{New pitch} = P \times (1 + 0.75) = P \times 1.75$$

Reduction in Number Divisions: The number of divisions is reduced by 50%. So, the new number of divisions is:

$$\text{New number of divisions} = N \times (1 - 0.5) = N \times 0.5$$

$$\text{New LC} = \frac{\text{New pitch}}{\text{New number of divisions}}$$

$$= \frac{P \times 1.75}{N \times 0.5}$$

$$\text{Substituting the values from the original least count:}$$

$$\text{New LC} = \frac{P \times 1.75}{N \times 0.5}$$

$$= 0.01 \text{ mm} \times 3.5 = 0.035 \text{ mm}$$

$$\text{New LC} = 35 \times 10^{-3} \text{ mm}$$

147. (b)  $(n+1) \text{ VSD} = n \text{ MSD}$

$$\Rightarrow 1 \text{ VSD} = \frac{n}{n+1} \text{ MSD}$$

$$\Rightarrow LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$LC = m - m \left( \frac{n}{n+1} \right) = m \left( \frac{n+1-n}{n+1} \right)$$

$$\Rightarrow LC = \left( \frac{m}{n+1} \right)$$

$$148. (b) \text{ Least count of the screw gauge}$$

$$= \frac{1 \text{ mm} - 0.01 \text{ mm}}{100}$$

$$\text{zero error} = +0.05 \text{ mm}$$

$$\text{Reading} = 4 \times 1 \text{ mm} + 60 \times 0.01 \text{ mm} - 0.05 = 4.55 \text{ mm}$$

149. (b) Least count of vernier calipers

$$= \frac{1}{20} \text{ cm}$$

Least count = 1 MSD - 1VSD

Let  $x$  no. of divisions of main scale coincides with  $N$  division of vernier scale, then

$$1 \text{ VSD} = \frac{x \times 1 \text{ mm}}{N}$$

$$\therefore \frac{1}{20} \text{ cm} = \frac{1 \text{ mm} - x \times 1 \text{ mm}}{N}$$

$$\frac{1}{20} \text{ mm} = 1 \text{ mm} - \frac{x}{N} \text{ mm} \Rightarrow x = \frac{2N-1}{2}$$

150. (d) Here, 9MSD = 10VSD and mass = 8.635 g

$$\text{For least count we have,}$$

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$LC = 1 \text{ MSD} - \frac{9}{10} \text{ MSD} = \frac{1}{10} \text{ MSD}$$

$$LC = 0.01 \text{ cm}$$

$$\text{Reading of diameter} = \text{MSR} + \text{LC} \times \text{VSR}$$

$$= 2 \text{ cm} + (0.01 \times 2) = 2.02 \text{ cm}$$

$$\text{Volume of sphere}$$

$$= \frac{4}{3} \pi \left( \frac{d}{2} \right)^3 = \frac{4}{3} \pi \left( \frac{2.02}{2} \right)^3 = 4.32 \text{ cm}^3$$

$$\text{Density}$$

$$= \frac{\text{mass}}{\text{volume}} = \frac{8.635}{4.32} = 1.998 - 2.00 \text{ g}$$

151. (c) 1VSD =  $\frac{49}{50}$  MSD

$$LC = \left( 1 - \frac{49}{50} \right) \text{ MSD} = \frac{1}{50} \text{ MSD}$$

$$\text{zero error} \Rightarrow +4 \times (LC) = 0.04 \text{ mm}$$

$$\Rightarrow 4 \times \frac{1}{50} \text{ MSD} = 0.04 \text{ mm}$$

$$\text{So, 1 MSD} = 0.05 \text{ mm}$$

$$1 \text{ cm} = \frac{1}{0.05} \text{ MSD}$$

$$1 \text{ cm} = 20 \text{ MSD}$$

152. (a) 1VSD =  $\frac{49}{50}$  MSD

$$= \frac{1}{50} \times \frac{1}{20} = 0.001 \text{ cm}$$

$$LC = 1 \text{ MSD} - 1 \text{ VSD} = 0.001 \text{ cm}$$

$$1 \text{ parsec} = 3.08 \times 10^{16} \text{ m}$$

$$1 \text{ Au} = 1.496 \times 10^{16} \text{ m}$$

$$\text{Therefore, Au} < h < \text{Parsec}$$

161. [220] Least count

$$= \frac{\text{Pitch}}{\text{No. of circular divisions}} = \frac{0.5 \text{ mm}}{100}$$

$$= 5 \times 10^{-3} \text{ mm}$$

$$\text{Positive Error} = \text{MSR} - \text{CSR} (LC)$$

$$= 0 \text{ mm} + 6.5 \times 10^{-3} \text{ mm}$$

$$\text{actual } L_1 = L_2 - L_3 = 71.61 - 40.51 = 31.10 \text{ mm}$$

$$\text{refractive index} = \frac{44.25}{31.10} = 1.42$$

$$153. (b) LC = \frac{\text{pitch}}{\text{No. of CSD}} = \left( \frac{1}{100} \right)$$

$$= 0.01 \text{ mm}$$

$$\text{Diameter} = \text{MSR} + \text{LC} \times \text{CSD}$$

$$\text{Diameter} = 1 + (0.01 \times 42 \text{ mm})$$

$$\text{Therefore, } \frac{x}{50} = 1.42$$

$$x = 71$$

154. (c) 20VSD = 19 MSD

$$\Rightarrow 1 \text{ VSD} = \frac{19}{20} \text{ MSD}$$

$$\text{given LC} = 0.01 \text{ mm} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$\Rightarrow 1 \text{ MSD} - 2 \text{ mm}$$

155. (d) Given: 10MSD = 11 VSD

$$\Rightarrow 1 \text{ VSD} = \frac{10}{11} \text{ MSD}$$

$$\text{As, LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{10}{11} \text{ MSD} = \frac{1 \text{ MSD}}{11}$$

$$= \frac{5}{11} \text{ units}$$

156. (d) 50 V = S = 495 - 5

$$S = 50 (S - V)$$

$$0.5 = 50 (S - V)$$

The Vernier constant of travelling microscope is

$$S - V = \frac{0.5}{50} = \frac{1}{100} = 0.01 \text{ mm}$$

157. (b) Spherometer is an instrument that can be used to measure curvature of surface.

158. (b)

159. (a) Zero Error =  $\pm$  Least Count

$$= \pm \text{Reading} + \text{MSR} - \text{VSR} - \text{LC} - \text{Zero Error}$$

Reading of Diameter = MSR + CSR (LC) - Positive zero error

$$= 4 \times 0.5 \text{ mm} + (46(5 \times 10^{-3})) - 6(5 \times 10^{-3}) \text{ mm}$$

$$= 2 \text{ mm} + 40 \times 5 \times 10^{-3} \text{ mm} = 2.2 \text{ mm}$$

162. (c) 1 M.S.D. = 1 mm

Now, 9 M.S.D. = 10 V.S.D.

$$1 \text{ V.S.D.} = 0.9 \text{ M.S.D.} = 0.9 \text{ mm} = 0.01 \text{ cm}$$

$$\text{Now, L.C. of vernier caliper} = 1 - 0.9 = 0.1 \text{ mm} = 0.01 \text{ cm}$$

$$\text{zero error} = -(10 - 4) \times 0.1 \text{ mm} = -0.6 \text{ mm}$$

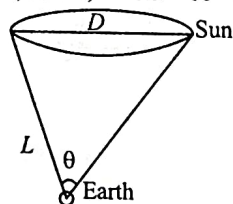
$$\text{Reading} = \text{M.S.R.} + \text{V.S.R.} - \text{Zero error}$$

$$= 3 \text{ cm} + 6 \times 0.01 - [-0.06]$$

$$= 3 + 0.06 + 0.06$$

$$= 3.12 \text{ cm} \sim \text{nearest to } 3.10$$

163. (c) Given, radius,  $r = 1.5 \times 10^{11} \text{ m}$



$$\theta = 2000 \text{ seconds } [\ell 1^\circ = 60', 1' = 60'']$$

$$= \frac{2000}{60 \times 60} \text{ degree} = \frac{2000}{60 \times 60} \times \frac{\pi}{180} \text{ radian}$$

$$= \frac{\pi}{36 \times 9} \text{ radian}$$

$$\text{Formula, Angle } (\theta) = \frac{\text{Arc}}{\text{radius}}$$

$$\theta = \frac{D}{r} \Rightarrow D = \theta \cdot r \quad [\text{arc} \approx D]$$

$$= \frac{\pi}{36 \times 9} \times 1.5 \times 10^{11} = 1.45 \times 10^9 \text{ m}$$

164. (c) MSD = 20 divisions per cm  $\Rightarrow 1$

$$\text{MSD} = \frac{1}{20} \text{ cm}$$

$$1 \text{ VSD} = \frac{24}{25} \text{ MSD}, \text{ L.C.} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{24}{25} \text{ MSD} = \frac{1}{25} \times \frac{1}{20} \text{ cm} = 0.002 \text{ cm}$$

165. (c) L.C. =  $\frac{0.5}{50} \text{ mm} = 0.01 \text{ mm}$

$$d = (2.5 + 45 \times 0.01) \text{ mm} = 2.95 \text{ mm}$$

$$\text{correct diameter} = d - \text{zero error}$$

$$= 2.95 - (-0.03) = 2.98 \text{ mm.}$$

166. [180] We have given, main scale reading = 1.7 cm

$$\text{Zero correction} = 0.05 \text{ cm}$$

$$\text{Vernier scale reading} = 5 \times 0.1 = 0.5 \text{ mm}$$

$$\text{Diameter of sphere} = \text{Main scale reading} + \text{Vernier scale reading} + \text{Zero correction}$$

$$= 1.7 + 0.05 + 0.05 = 1.8 \text{ cm} = 180 \times 10^{-2} \text{ cm.}$$

167. [412] 1MSD = 1mm

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$\text{Therefore, } 1 \text{ VSD} = 0.9 \text{ MSD}$$

$$\text{L.C} = 1 \text{ MSD} - 1 \text{ VSD} = 1 - 0.9 = 0.1 \text{ mm}$$

$$\text{Positive zero error} = 4 \text{ LC} = 0.4 \text{ mm}$$

$$\text{Reading} = \text{MSR} + \text{VSR} + \text{correction}$$

$$= 4.1 \text{ cm} + 6 \times 0.01 \text{ cm} + (-0.04 \text{ cm})$$

$$= 4.12 \text{ cm} = 412 \times 10^{-2} \text{ cm}$$

168. [5] Let  $x$  be the value of one division of main scale

$$\therefore x = \frac{1}{20} \text{ cm} = 0.05 \text{ cm}$$

Let  $y$  be value of one division on vernier

$$\text{scale } 10y = 9x \Rightarrow y = \frac{9x}{10}$$

$$\text{Least count} = x - \frac{9x}{10} = \frac{x}{10} = \frac{0.05}{10}$$

$$= 0.005 \text{ cm} = 5 \times 10^{-2} \text{ mm}$$

169. (b) L.C. =  $\frac{\text{Pitch}}{\text{No. of division on circular scale}}$

$$= \frac{0.5}{50} \text{ mm} = 0.01 \text{ mm}$$

$$\text{Diameter, } d = 1.5 + 7 \times 0.01$$

$$= 1.57 \text{ mm} = 0.157 \text{ cm}$$

$$\therefore \text{Curved Surface Area} = (2\pi r)\ell = \pi d\ell$$

$$(\ell = d = 2r)$$

$$= 3.142 \times 0.157 \times 6.8 = 3.354 \text{ cm}^2 = 3.4 \text{ cm}^2$$

170. (b) Number of divisions = 50, Pitch = 0.5 mm

$$\text{L.C.} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

$$\text{True Reading} = 5 \text{ mm} + 20 \times 0.01 - 5 \times 0.01 = 5.15 \text{ mm}$$

171. (b) Reading = MSR + VSD  $\times$  LC - zero error

$$\text{Reading} = 8.5 + \frac{(0.1) \times 6}{10} - \frac{0.2}{10} = 8.54 \text{ cm}$$

172. [25]  $4t = 100 \times \frac{4}{3} \pi \times \frac{3}{40\pi} \times 10^{-9} \times 0.01$

$$\Rightarrow t = 2.5 \times 10^{-11} \text{ cm} = 25 \times 10^{-14} \text{ m.}$$

173. (a) LC =  $\frac{\text{Pitch}}{\text{No. of division}} = \frac{1}{100} \text{ mm}$

$$= 0.01 \text{ mm}$$

$$\text{Zero error} = +0.08 \text{ mm.}$$

$$\text{Diameter} = 1 + 72 \times 0.01 - 0.08 = 1.64 \text{ mm}$$

$$\text{Radius} = 0.82 \text{ mm}$$

174. (b) Least count

$$= \frac{\text{Pitch}}{\text{Total divisions on circular scale}}$$

In 5 revolution, distance travelled = 5 mm  
 $\therefore$  In 1 revolution, distance travelled = 1 mm

$$\text{L.C.} = \frac{1}{50} = 0.02 \text{ mm} = 0.002 \text{ cm}$$

Therefore, Assertion is not correct but reason is correct

175. (c)  $(n - 1) \text{ MSD} = n \text{ VSD}$

$$\Rightarrow \text{VSD} = \frac{n-1}{n} (\text{MSD})$$

$$\text{Now, L.C.} = [1 \text{ MSD} - 1 \text{ VSD}] \text{ cm}$$

$$= a - \frac{n-1}{n} a = \frac{a}{n} \text{ cm} = \left( \frac{10a}{n} \right) \text{ mm}$$

176. [52] Given: 9 MSD = 10 VSD

$$\text{Least count of vernier} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{9}{10} \text{ MSD}$$

$$\text{MSD} = 0.1 \text{ MSD} = 0.1 \times 1 \text{ mm} = 0.1 \text{ mm}$$

$$\text{Reading} = 10 \text{ mm} + 8 \times 0.1 \text{ mm} = 10.8 \text{ mm}$$

$$\text{Final reading} = 10.8 \text{ mm} - 0.04 \text{ mm} = 10.4 \text{ mm}$$

$$\text{So, radius of bob} = 5.2 \text{ mm} = 52 \times 10^{-2} \text{ cm}$$

177. [13] Difference in Reading

$$= \text{Positive Zero Error} - \text{Negative Zero Error}$$

$$= (+5) - [-(100 - 92)] = 13$$

178. (b) Least count =  $\frac{0.5 \text{ mm}}{50}$

$$= 1 \times 10^{-5} \text{ m} = 10 \mu\text{m}$$

179. (\*)  $A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2$

$$= 280.6324$$

$$A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2$$

$$= 280.722$$

$$A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183$$

$$= 280.6642$$

$$A_4 + B_4 + C_4 = 25 + 236.191 + 19.5$$

$$= 280.691$$

So, no option is correct. (Bonus)

180. (c) Least count =  $\frac{P}{N} = \frac{0.1 \text{ cm}}{50} = 0.002 \text{ cm}$

$$\text{mm} = 0.002 \text{ cm}$$

Hence, measurement should be integral multiple of L.C.

181. (b) Least count

$$= \frac{\text{Pitch}}{\text{No. of divisions on circular scale}}$$

$$\Rightarrow 5 \times 10^{-6} = \frac{10^{-3}}{N} \Rightarrow N = 200$$